Progress Report: Non-Parametric Rank-Shift Selectivity and Cross-Method Synthesis (Next Steps #3)

Author: Vladislav Mantsevich
Affiliation: Utrecht University
Project: EV Uptake Proteomics (PC3 vs. PC3M)
Date: November 12, 2026

1. Overview

Standard differential uptake analysis using parametric linear models (limma-trend) evaluates continuous fold changes of normalized EV cargo. However, intensity-based parametric models can be vulnerable to normalization artifacts and variance inflation near the analytical noise floor. In our quantitative setup, after technical curation, the functional uptake cohort is constrained to six validated biological channels: $n_1 = 2$ replicates for recipient osteoblasts (SAOS-2) incubated with parental PC3 EVs, and $n_2 = 4$ replicates for SAOS-2 incubated with metastatic PC3M EVs (channel PC3_EV_R1 was excluded due to library preparation failure). Under such sample-constrained conditions, standard machine learning classification is unfeasible due to high risk of overfitting and insufficient degrees of freedom.

To provide an orthogonal validation layer, we implemented a non-parametric rank-shift evaluation across the quantified EV cargo proteome. If parametric modeling assumptions are biologically valid, relative fold changes should strongly correlate with ordinal rank promotions between unincubated baseline mixtures and cellular uptake channels.

The framework directly contrasts two experimental states across multiplexed channels:

- **Spike-In Reference Channels:** Recipient SAOS-2 cell lysate mixed with donor EVs without incubation, capturing baseline cargo abundance, protein yield, and ionization characteristics prior to uptake.
- **Natural Uptake Channels:** Recipient SAOS-2 cells incubated with donor EVs to allow physiological internalization.

Within each individual TMT channel, proteins are ranked by abundance from lowest (1) to highest (N), splitting the difference for tied values.

2. Internalization Selectivity within Phenotypes ($\Delta\mathrm{Rank}$)

To account for biological and technical variability across replicates, ranks were averaged within each experimental condition:

$$\mathrm{Rank}_{\text{Spike-in}} = \mathrm{mean}(\text{Spike-in Ranks}), \quad \mathrm{Rank}_{\text{Uptake}} = \mathrm{mean}(\text{Uptake Ranks})$$

The directional shift in proteomic hierarchy defines the internalization selectivity ($\Delta\mathrm{Rank}$):

$$\Delta\mathrm{Rank} = \mathrm{Rank}_{\text{Uptake}} - \mathrm{Rank}_{\text{Spike-in}}$$

The sign and magnitude of $\Delta\mathrm{Rank}$ classify cargo sorting behavior in recipient osteoblasts:

- **$\Delta\text{Rank} \approx 0$ (Neutral):** The protein maintains its relative rank within the proteome hierarchy, showing no detectable bias relative to the baseline mixture.
- **$\Delta\text{Rank} > 0$ (Relative Enrichment):** The protein advances in rank following incubation, indicating net over-representation in recipient cells. While this shift is consistent with processes such as preferential uptake, surface retention, or intracellular stability, the precise underlying mechanisms remain undetermined.
- **$\Delta\text{Rank} < 0$ (Relative Depletion):** The protein drops in rank following incubation, indicating net under-representation in recipient cells. This pattern may reflect inefficient entry, reduced persistence, or accelerated turnover, though the specific cause is currently unknown.

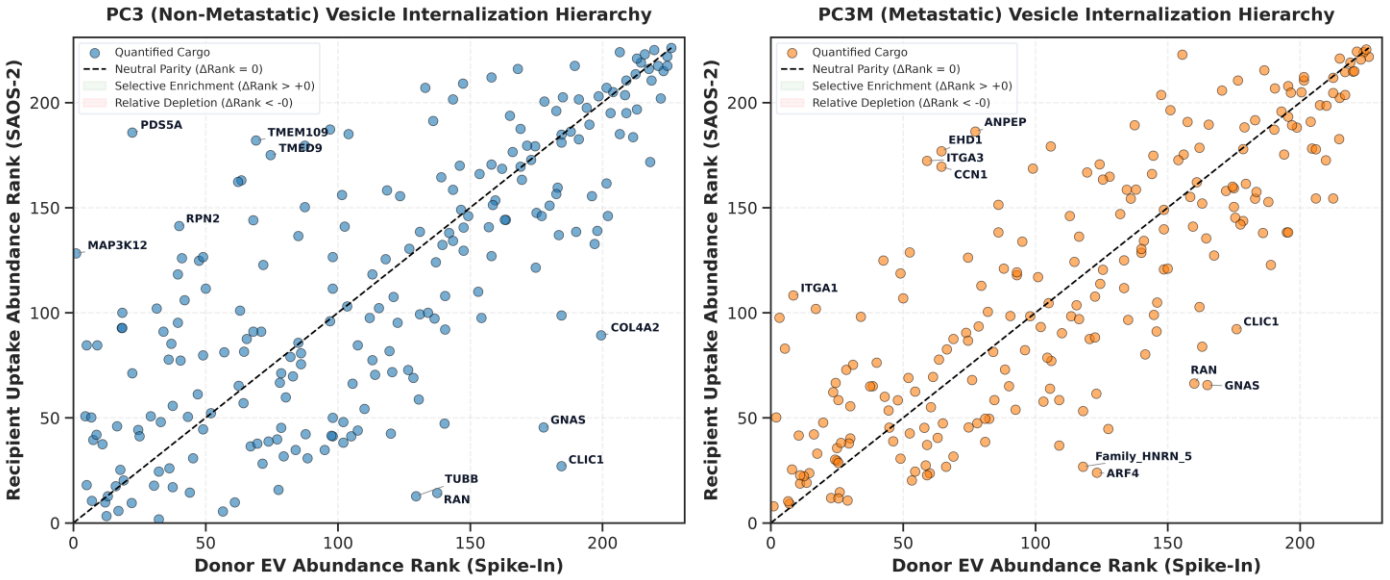


Figure 1. Protein internalization hierarchies for parental PC3 (left) and metastatic PC3M (right) vesicles in recipient osteoblasts. The dashed diagonal line denotes neutral parity ($\Delta\text{Rank} = 0$), where cargo entry follows non-selective mass action. Points deviating markedly from the parity line highlight selective cargo sorting: proteins positioned well above the diagonal represent preferential internalization or intracellular accumulation in SAOS-2 cells, whereas proteins lying well below it indicate relative depletion or inefficient entry. Key divergent drivers are highlighted with gene labels.

3. Cross-Phenotype Selectivity Contrast ($\Delta\Delta\text{Rank}$)

While ΔRank identifies active uptake within each vesicle lineage separately, $\Delta\Delta\text{Rank}$ was established to determine whether internalization dynamics in osteoblasts genuinely differ between parental (PC3) and metastatic (PC3M) vesicle pools:

$$\Delta\Delta\text{Rank} = \Delta\text{Rank}_{\text{PC3M}} - \Delta\text{Rank}_{\text{PC3}}$$

Expanding the terms into their underlying experimental states:

$$\Delta\Delta\text{Rank} = \left(\text{Rank}_{\text{PC3M, Uptake}} - \text{Rank}_{\text{PC3, Uptake}}\right) - \left(\text{Rank}_{\text{PC3M, Spike-in}} - \text{Rank}_{\text{PC3, Spike-in}}\right)$$

Subtracting baseline spike-in differences cancels out pre-existing abundance disparities between donor vesicle preparations (e.g. if a protein is simply 10-fold more abundant in PC3M vesicles and enters osteoblasts via non-selective bulk endocytosis, its uptake rank and spike-in rank shift equally, resulting in $\Delta\Delta\text{Rank} \approx 0$).

Consequently, non-zero values isolate phenotype-specific internalization preference:

- **$\Delta\Delta\text{Rank} > 0$:** Recipient osteoblasts preferentially internalize or retain this cargo from metastatic PC3M vesicles over parental PC3 vesicles, exceeding initial donor dosage.
- **$\Delta\Delta\text{Rank} < 0$:** Internalization is selectively favored for cargo delivered by parental PC3 vesicles.

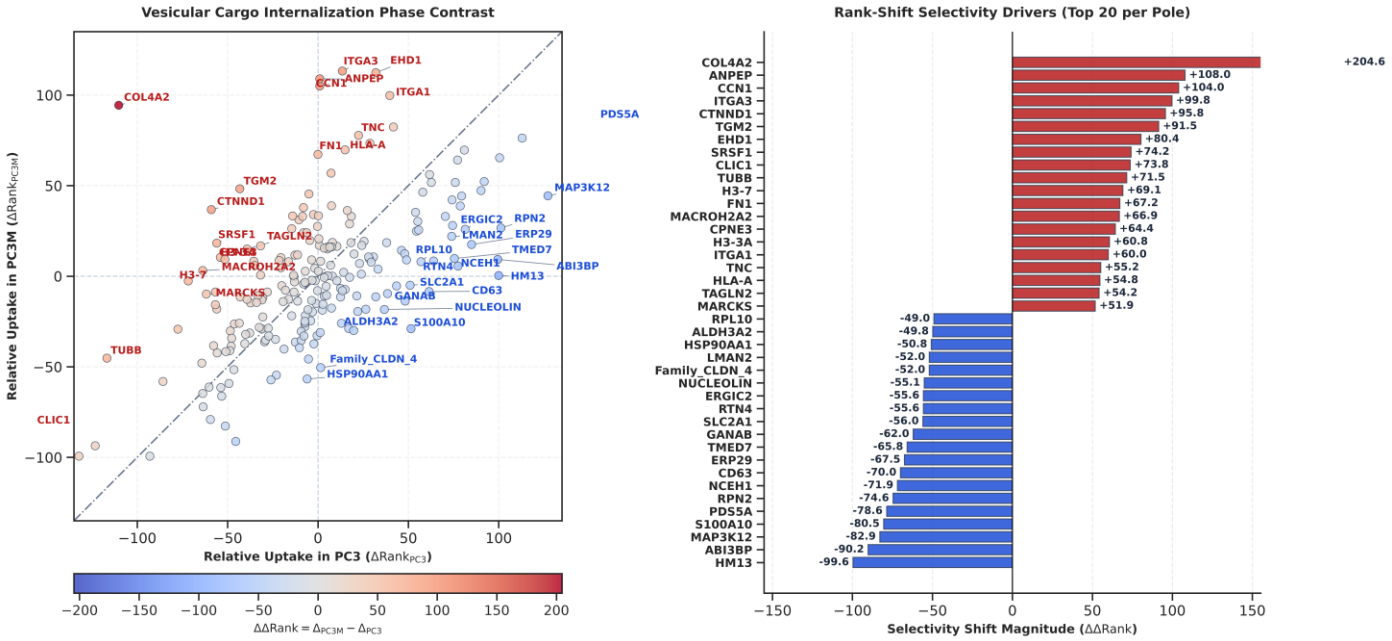


Figure 2. Internalization Phase Contrast and Top Selectivity Drivers.
Left panel (Phase Contrast): Plots the rank shift observed under metastatic vesicle uptake (ΔRank_{PC3M}, y-axis) against the rank shift under parental vesicle uptake (ΔRank_{PC3}, x-axis). The diagonal identity line ($y = x$) represents identical cargo sorting efficiency across both phenotypes. Divergent proteins fall away from the diagonal and are colored by their $\Delta\Delta$Rank magnitude (red: metastatic PC3M bias; blue: parental PC3 bias).
Right panel (Diverging Barplot): Displays the top 20 selective cargo drivers for each phenotype. Red bars show proteins most strongly enriched in metastatic PC3M uptake ($\Delta\Delta\text{Rank} > 0$), while blue bars highlight proteins favored under parental PC3 conditions ($\Delta\Delta\text{Rank} < 0$).

4. Method Benchmarking: Linear Model Validation via Rank Shifts

Having derived the non-parametric $\Delta\Delta\text{Rank}$ metric in the previous section, our main goal was to test whether our primary parametric framework (limma-trend) holds ground, rather than establishing an alternative selection pipeline. Because rank analysis operates on raw ordinal hierarchies, it serves as an assumption-free check to verify that our limma results reflect robust directional trends rather than data-processing artifacts.

Across the entire dataset, continuous relative fold changes ($\log_2 FC$) and rank shifts ($\Delta\Delta\text{Rank}$) show an exceptionally strong Spearman correlation ($r_s = 0.91$, Figure 3B). This near-linear alignment confirms that the parametric model mirrors the physical promotion and demotion of proteins within the proteome hierarchy.

To visualize this agreement in discrete terms, we applied an exploratory rank cutoff of $|\Delta\Delta\text{Rank}| \geq 50.0$ (concordant sign) alongside our standard limma thresholds ($\text{FDR} < 0.10$, $|\log_2 FC| \geq 1.0$) (Figure 3).

Despite the rank threshold of 50 being a purely empirical, exploratory cutoff chosen for visualization, the two methods show substantial overlap (27 proteins meet both criteria, with only 15 unique to limma and 12 to rank shifts; Figure 3A).

This non-parametric exercise served strictly as a macro-level validation. Because rank shifts lack formal error modeling, variance moderation, and false-discovery rate control, ordinal ranks are not utilized for downstream feature selection.

With the primary linear model independently benchmarked and confirmed by this high rank correlation, we conclude the non-parametric evaluation here. All subsequent biological analyses, functional enrichments, and mechanistic conclusions rely entirely on the statistically significant candidates authenticated by Empirical Bayes moderation (limma).

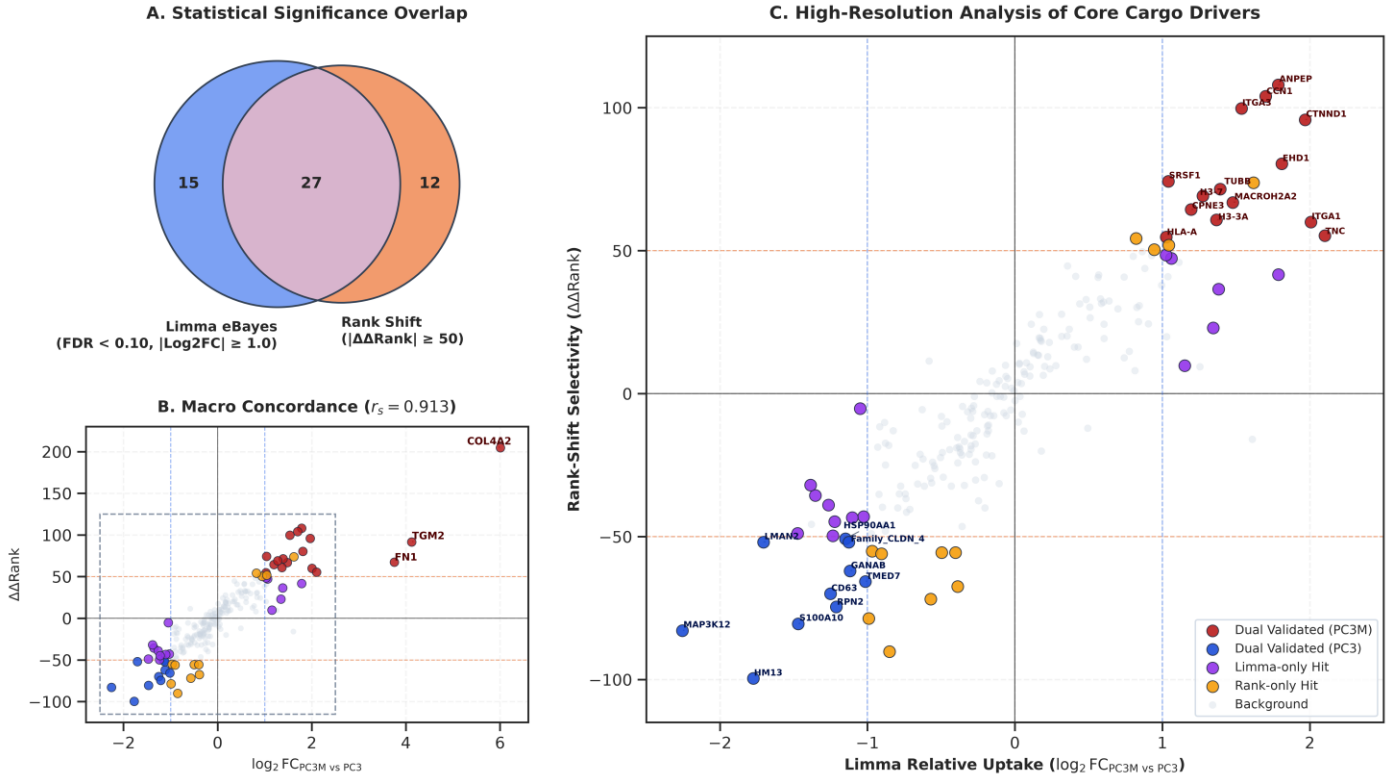


Figure 3. Multi-Method Concordance Landscape and Core Cargo Drivers.
(A) Overlap of parametric and non-parametric hits: Venn diagram illustrating concordance between targets identified by empirical Bayes linear modeling (limma-trend: FDR < 0.10, $|\log_2 \mathrm{FC}| \geq 1.0$) and rank-shift analysis ($|\Delta\Delta\mathrm{Rank}| \geq 50.0$, matching sign). In total, 27 core cargo proteins fulfill both criteria simultaneously, whereas 15 are detected exclusively by limma and 12 exclusively by rank shift.
(B) Global concordance across the proteome: Scatter plot of relative differential uptake ($\log_2 \mathrm{FC}$) versus $\Delta\Delta\mathrm{Rank}$ across all quantified proteins exhibiting a high Spearman rank correlation ($r_s = 0.91$). This tight agreement demonstrates that continuous intensity gains correspond directly to promotions within the physical abundance hierarchy.
(C) High-resolution concordance landscape: Four-quadrant scatter plot mapping continuous effect sizes ($\log_2 \mathrm{FC}$) against rank shifts ($\Delta\Delta\mathrm{Rank}$). Targets in the upper-right and lower-left quadrants demonstrate concordant directional behavior: core drivers enriched in metastatic PC3M uptake display high positive values along both axes, while parental PC3-selective cargo segregates into the negative domain. Points in the off-diagonal areas mark method-specific edge cases, illustrating why parametric statistical moderation remains the decisive criterion for downstream candidate selection.